

Please write clearly in block capitals.

Centre number

Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE**Paper 1 (Generated practice paper - Set 24)**

Assessed: The physical environment, Energy resources, Pollution, Research methods

Time allowed: 3 hours**Materials**

For this paper you may use:

- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
 - use good English
 - organise information clearly
 - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
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10	
11	
TOTAL	

Answer all questions in the spaces provided.

0 1

Table 1 shows the mean surface temperature of the Earth with and without an atmosphere containing greenhouse gases.

Table 1

Condition	Mean surface temperature / °C
No greenhouse effect	-18
Natural greenhouse effect	15

0 1

. 1

Use Table 1 to calculate the warming produced by the natural greenhouse effect.

[1 mark]

Answer

°C

0 1

. 2

Explain how the natural greenhouse effect warms the Earth's surface.

[4 marks]

0 1

. 3

Explain what is meant by a **greenhouse gas**.

[1 mark]

0 1 . 4

The wavelengths of radiation arriving at the Earth are different from those leaving. Describe this difference and explain why it matters to the energy budget.

[2 marks]

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0 1 . 5

Explain why a change in the movement of energy through the atmosphere could affect the distribution of species.

[2 marks]

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10

0 2 . 1

Describe how human activities have altered the natural equilibria of the carbon and nitrogen cycles, and evaluate the strategies that could counteract these changes.

[9 marks]

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0 2 . 2

State what is meant by the **residence time** of a substance in a reservoir.**[1 mark]**

10

0 3 . 1

Which of the following water sources is usually the most convenient for large communities? Tick one box.

[1 mark]

A Desalinated seawater

☐

B River water

☐

C Rainwater catchment

☐

D Newly discovered aquifers

☐

0 3 . 2

Explain why the quality requirements for water differ between uses, using **two** named uses.**[2 marks]**

0 3 . 3

Explain why inter-basin transfers may be used and give **one** disadvantage.**[2 marks]**

5

04 . 1

Which chemical form of mercury is most easily absorbed and biomagnified in food chains? Tick **one** box.

[1 mark]

- A Elemental (liquid) mercury
 B Inorganic mercury compounds
 C Organic (methyl) mercury
 D Mercury vapour

☐☐☐☐

04 . 2

Explain why anaerobic sediments increased the hazard of the inorganic mercury discharged at Minamata.

[2 marks]

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04 . 3

Explain why heavy metal wastes are stored under alkaline conditions.

[2 marks]

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05 . 1

Describe coal gasification and coal liquefaction and explain how each increases the usefulness of coal.

[4 marks]

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0 5 . 2 Describe what methane hydrate is and where it is found.

[2 marks]

0 5 . 3 Describe **two** proposed methods of extracting methane from hydrate deposits and explain **two** environmental risks.

[4 marks]

10

0 6 Table 1 shows the running mean of moth catches as a student increased the number of nights of light-trapping at one site.

Table 1

Number of nights	1	2	4	6	8	10	12	14
Running mean catch per night	31	18	24	21	22	22	21	22

0 6 . 1 Use Table 1 to suggest the minimum number of nights needed to obtain a reliable mean. Explain your answer.

[2 marks]

06 . 2

Explain why moth surveys must be carried out at standardised times and weather conditions.

[3 marks]

06 . 3

Distinguish between **accuracy** and **precision** of measurements.

[2 marks]

06 . 4

Explain the importance of a null hypothesis and standardised techniques in a scientific investigation.

[3 marks]

10

07 . 1

Explain why current global energy use is described as unsustainable.

[3 marks]

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07 . 2

Describe the environmental impacts of exploiting fossil fuels that occur **before** the fuel is used.

[4 marks]

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07 . 3

Explain why an 'energy gap' could occur if society waits until fossil fuels become expensive before developing alternatives.

[3 marks]

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08

Figure 1 shows the number of breeding pairs of sparrowhawks in a study area in the UK between 1945 and 1985, and the years in which DDT use began and was banned.

Figure 1



08

1

Describe the trend shown in Figure 1.

[2 marks]

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08

2

Calculate the percentage decrease in breeding pairs between 1950 and 1965. Show your working.

[2 marks]

Answer

 %

0	8	.	3
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DDT use in the UK began in the late 1940s and was restricted from the mid 1960s. Explain how the properties of DDT could account for the pattern in Figure 1.

[4 marks]

0	8	.	4
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Suggest why the recovery after 1970 was slow.

[2 marks]

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0	9	.	1
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Explain how hydrogen can be used to store energy from intermittent renewable resources, and describe **two** ways the stored energy can be released.

[4 marks]

0 9 . 2 Compare **three** methods of storing hydrogen, giving a disadvantage of each. **[3 marks]**

0 9 . 3 Explain how a fuel cell differs from a rechargeable battery. **[2 marks]**

0 9 . 4 Explain what is meant by the 'hydrogen economy'. **[1 mark]**

10

1 0 Table 1 shows the mean daily solar energy received on a horizontal surface at three locations.

Table 1

Location	Latitude	Mean solar energy in June / kWh m ⁻² day ⁻¹	Mean solar energy in December / kWh m ⁻² day ⁻¹
Sahara desert	25°N	7.8	4.2
Southern England	51°N	5.0	0.7
Northern Norway	69°N	4.6	0.0

1 0 . 1 Use Table 1 to calculate how many times more solar energy is received in December in the Sahara than in southern England. **[1 mark]**

Answer

times

1 0 . 2

Explain why northern Norway receives more solar energy in June than in December.

[2 marks]

1 0 . 3

Explain why concentrating solar power systems using parabolic reflectors only work where there is no cloud.

[2 marks]

1 0 . 4

Explain why the low energy density of solar power is a problem for large-scale generation.

[1 mark]

1 0 . 5

Evaluate the extent to which new technologies could overcome the problems of using solar power as a large-scale energy resource.

[9 marks]

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Extra space

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15

Write an essay on one of the following topics.

$$\boxed{1} \boxed{1} . \boxed{1}$$

Discuss how the properties of solar power and hydroelectric power affect their use, and evaluate how new technologies could increase their contribution to energy supplies.

[25 marks]

OR

1	1	.	2
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Evaluate the contribution of new technologies to the monitoring and understanding of environmental change in the physical environment.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS